Worksheet 8.1

Properties of materials

Student: ..... Class: .....

Natural or processed?

A ‘natural’ material is a substance that exists in nature, while a ‘processed’ material is one that has been altered, created or manufactured by human intervention.

Consider the following substances/objects:

steel	water	wood	glass
baby	plastic	gold	wheat
paper	breakfast cereal	air	alcohol

1. Enter each of these into one of the two columns of the table below according to whether you consider them to be natural or processed.

Natural	Processed

2. Were there any substances/objects in this list that could have been placed in either column? If so what were they?

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3. While milk is produced naturally by cows, the milk that we buy in the supermarket has been subjected to a number of different processes such as pasteurisation (which sterilises it) and homogenisation (which mixes the cream and the milk together uniformly). Would you describe milk as natural or processed? Explain your answer.

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Useful properties

The properties of a material are the physical characteristics that the material has, and will determine the use that it is put to. For example, glass is used to make windows because it is transparent. Transparency is one of the properties of glass.

1. For each of the following materials, state what property of the material makes it useful for the stated purpose. The first one has been done for you.

Material	Used to make…	Useful property of material
Glass	Windows	Transparency
Styrofoam	Disposable coffee cups	
Steel	Bridges	
Gold	Jewellery	
Aluminium	Saucepans	
Foam rubber	Pillows	
Sugar	Lollies	
Wood	Matchsticks	
Plastic	Shopping carrier bags	
Elastic	Hair bands	

2. Give reasons why you *wouldn't* use:

(a) plastic to make a frying pan

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(b) wood to make pillows

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(c) rubber to make a bridge

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(d) sugar to make jewellery

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(e) elastic to make matchsticks.

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